Sleep, Screen Time and Stress Analysis

Modern lifestyle habits, particularly **excessive smartphone usage**, are frequently cited as disruptors of circadian rhythms and mental health. However, the specific "tipping point" where usage becomes detrimental to sleep quality and stress levels is often unclear for different demographic groups.

1. Problem Statement Formation

"To what extent does daily **smartphone usage duration** correlate with decreased **sleep quality** and increased **mental fatigue**, and how do lifestyle factors like **physical activity** and **caffeine intake** mediate these effects across different age groups?"

2. Context

As digital integration increases, "Technostress" has become a significant public health concern. This project seeks to move beyond the "phones are bad" narrative to find **actionable patterns**—such as whether high physical activity can offset the negative impact of screen time.

3. Criteria for Success

- **Statistical Significance:** Identify at least three key predictors of "Poor Sleep Quality" with a p-value > 0.05.
- **Predictive Accuracy:** Build a classification model (e.g., Random Forest or Neural Network) that can predict "High Stress" or "Poor Sleep" with an **F1-score of 0.75 or higher**.
- **Actionable Insights:** Define specific "Usage Thresholds" (e.g., *"Users over 30 with >5 hours of usage see a 20% drop in sleep quality"*).

5. Scope of Solution Space

- **In-Scope:** Correlation analysis between usage and sleep; demographic segmentation; impact of lifestyle "buffers" (exercise).
- **Out-of-Scope:** Longitudinal tracking; medical diagnosis of sleep disorders; clinical mental health interventions.

6. Constraints

- **Self-Reporting Bias:** Recognition that "Sleep Quality" and "Stress" are subjective measures and may vary based on individual perception.
- **Causality:** The analysis can show **correlation**, but cannot definitively prove that phones *cause* stress (it could be that stressed people use phones more).
- **Data Privacy:** All demographic data must be handled according to **GDPR/HIPAA** standards (anonymized).

7. Stakeholders

- **Health & Wellness App Developers:** To build better "Digital Detox" features.
- **Public Health Researchers:** To inform guidelines on healthy device usage.
- **The End User:** To receive personalized feedback on how their habits affect their rest.

8. Data Sources

- **Primary synthetic dataset:** "Sleep, Screen Time and Stress Analysis" found on Keggle is a great way of exploring the intersection of technology and well-being. It contains15,000 records representing individuals including:
    - **Features:** Age, Gender, Occupation, Usage Patterns, Caffeine, Activity.
    - **Targets:** Sleep Quality, Sleep Duration, Stress Level, Mental Fatigue Score.